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Comparison of eccentric hamstring strength and asymmetry at return-to-sport after hamstring strain injury among those who did and did not re-injure

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ABSTRACT

Objectives: The primary aim of this study was to describe eccentric hamstring strength magnitude and asymmetry at the time of return-to-sport (RTS) after an index hamstring strain injury (HSI) and determine if there were differences in strength asymmetry at RTS between those who did and did not go on to re-injure within 1-month and within 3-months of RTS.
Design: Cross-sectional study.
Setting: Laboratory-based.
Participants: Sixty National Collegiate Athletic Association Division I athletes with index HSI. Main Outcome Measures: Maximum hamstring eccentric strength for each limb, total maximum strength summed across limbs, and between-limb asymmetry at the time of RTS following the index HSI, assessed using the NordBord Hamstring Testing System.
Results: Of the 60 index HSIs, 8 (13%), and 11 (18%) re-injuries occurred within 1 and 3-months of RTS, respectively. There were no differences between those who did and did not re-injure in maximum eccentric force of either limb (p-values≥0.52), total force from both limbs (p-values≥0.47), and between limb force asymmetry (p-values≥0.91), regardless if re-injury occurred within 1 or 3-months after RTS.
Conclusions: Eccentric hamstring strength and asymmetry measured at the time of RTS did not differ between those who did and did not re-injure within 3-months of RTS.
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1. Introduction

Hamstring strain injuries (HSI) account for over one-third of injuries in sports involving sprinting and typically require several weeks away from participation (Ekstrand et al., 2012; Jones et al., 2019; Roe et al., 2018a; Silvers-Granelli et al., 2021), resulting in missed competitions and potentially reduced performance upon return (Ekstrand et al., 2016; Hallén & Ekstrand, 2014; Roe et al., 2018b). Despite the efforts to prevent HSI, the injury incidence (between 1.06/1000 and 5.87/1000 h) and re-injury rate (14–63%) remain high, especially in athletes such as American football, soccer, and track athletes (De Visser et al., 2012; Ekstrand et al., 2011, 2013; Wille et al., 2022). While prior HSI and age are generally accepted risk factors, the predictive capabilities of other parameters, such as eccentric strength deficits and monitorization timing, are debated (Baroni & Oliveira Pena Costa, 2021; Green et al., 2020; Hickey et al., 2017). Additionally, it has been shown that more than 50% of re-injuries occur within the first 4 weeks after RTS and more than 70% of re-injuries occur within the first 100 days for soccer and American football (Bodendorfer et al., 2021; Wangensteen et al., 2016). However, the time frame used to define re-injury in prior research ranges from 1 to 12 months (Bodendorfer et al., 2021; Ekstrand et al., 2012; Heiderscheit et al., 2022; Wangensteen et al., 2016), further contributing to inconsistent findings across studies to this point.

One HSI mechanism occurs when the hamstrings are lengthening while producing force (eccentric) to slow the limb during the swing phase of high-speed running (Chumanov et al., 2011; Heiderscheit et al., 2005; Hickey et al., 2022; Kenneally-Dabrowski et al., 2019). For this reason, eccentric strength assessments are

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frequently used as targets for predicting HSI injury risk and determining an athlete's readiness for return to sport (RTS) (Bourne et al., 2015; Opar et al., 2015; Timmins et al., 2016; Wille et al., 2022). While initial studies suggested that preseason eccentric hamstring strength deficits were predictive of index HSI (Opar et al., 2015; Timmins et al., 2016), more recent studies, including a meta-analysis (Opar et al., 2021), found no relationship between strength and index HSI occurrence (Van Dyk et al., 2017; Wille et al., 2022). However, up to a 10% concentric strength deficit or lower strength compared to uninjured athletes has been observed at the time of RTS, which is thought to be related to the remaining muscle injury observed on MRI (de Lima-E-Silva et al., 2020; Ribeiro-Alvares et al., 2021; Sanfilippo et al., 2013). These persistent strength deficits at the time of RTS following an index HSI may be associated with increased re-injury (Green et al., 2020).

Therefore, the primary aim of this study was to measure eccentric hamstring strength and asymmetry at the time of RTS after an index HSI (it refers to the first HSI within the surveillance period) and determine if there were differences in strength asymmetry at RTS between those who did and did not go on to re-injure within 1 month and within 3 months. We hypothesized that eccentric hamstring strength asymmetry at RTS would be greater in athletes who went on to re-injure compared to those who did not. The secondary aim was to describe eccentric hamstring strength asymmetry and re-injury occurrence according to the index HSI injury severity, as measured by days to RTS.

2. Methods

2.1. Study design

Routinely collected health and performance data from National Collegiate Athletic Association Division I athletes in the University of Wisconsin-Madison Badger Athletic Performance database were used in this study. HSI occurrence was collected as part of a larger prospective, observational study monitoring acute HSIs in collegiate athletes (Wille et al., 2022). The University's Health Sciences Institutional Review Board approved all study procedures and athletes provided written informed consent.

2.2. Participants

Records of male and female athletes participating in American football (male only), soccer, and track who sustained an HSI between July 2017 and March 2020 were reviewed and extracted if the following criteria were met: the athlete 1) was healthy and cleared for full athletic participation at the time of testing; and 2) did not have an injury that would limit sports participation during the 3 months prior to the index HSI. An HSI was defined as a sudden onset of posterior thigh pain occurring during training or game, causing the athlete to miss at least one training session or game, and the presence on physical examination of two or more of the following: 1) pain with palpation of the hamstring muscles, 2) posterior thigh pain during straight leg raise, 3) weakness with resisted knee flexion, and 4) pain with resisted knee flexion (Heiderscheit et al., 2010). The use of “index HSI” was used for HSIs that occurred after no hamstring injury in the last 3 months during the surveillance period. All HSIs were diagnosed by a member of the respective teams' sports medicine staff and tracked by licensed athletic trainers. For each eligible athlete record, hamstring eccentric strength at the time of RTS following the index HSI was extracted, in addition to age, height, mass, gender, sport, injured limb, days from index HSI to RTS, and subsequent HSI during a 3-month period after RTS. All athletes followed the standard rehabilitation protocol after HSI, guided by the team's sports medicine staff.

2.3. Procedures

Eccentric hamstring strength was measured as previously described using the NordBord Hamstring Testing System (Vald Performance, Newstead QLD, Australia) (Fig. 1). (Opar et al., 2013) Briefly, athletes kneeled on a padded platform with hooks around the ankles just above the lateral malleolus. Athletes were instructed to descend as slowly as possible, with the hands crossed on the trunk, and the shoulders, hips and knees in a straight line. Three warm-up trials were completed, one each at 50%, 75%, and 90% of maximal effort. After the warm-up, 3 maximum effort trials were completed with verbal encouragement and a 30-s rest interval between trials. If there was more than 20% difference between the trials’ peak forces or if the athlete was unable to complete the trial with proper alignment (invalid trials), additional trials were performed up to a maximum of 5 trials. All valid trials were recorded and the peak force for each limb across all trials was used for analysis.

The number of days from index HSI to RTS was used as an indicator of injury severity and was classified in subcategories as minimal (1–3 days), mild (4–7 days), moderate (8–28 days), severe (>28 days) (Fuller et al., 2006). Following RTS, all athlete re-injuries were tracked by the team athletic trainers. Re-injuries were defined as an acute HSI to the same limb as the index HSI within the 3-month period after RTS. Any HSIs involving the contralateral limb, relative to the index HSI, were not included (Wangensteen et al., 2016).

2.4. Statistical analyses

Eccentric hamstring strength was assessed by maximum force

Fig. 1. The starting position for eccentric hamstring strength testing on the NordBord Hamstring Testing System.

for each limb, total maximum force, and maximum force asymmetry between limbs. Maximum forces of the injured limb ($F_{injured}$) and non-injured limb ($F_{non\text{-}injured}$) were determined for each athlete across all trials and summed to calculate total force ($F_{injured} + F_{non\text{-}injured}$). The between limb asymmetries in peak forces were calculated using equation $|(F_{non\text{-}injured} - F_{injured})|/F_{non\text{-}injured}$. Peak forces were also normalized by body mass (N/kg).

Data analysis was performed using the Statistical Package for the Social Sciences (SPSS Inc., version 21; IBM, Raleigh, NC) for Windows. Kolomogorov-Smirnov test was used for determining the normality of the distribution of variables. Normality analysis showed that the eccentric strength data were not normally distributed. Therefore, count data were summarized as N (%) and continuous data were expressed as median and interquartile range (IQR). Mann-Whitney *U* test was used to examine between-group differences. The p-values were deemed significant at <0.05. X.

3. Results

Data from 60 unique athletes (11 female) from three different sports (football, track, soccer) were included in the study. Athlete characteristics at the time of the index HSI are presented in Table 1.

Eccentric hamstring strength values between athletes with and without re-injury in the first 1 month and first 3 months after RTS are given in Table 2. Of the 60 index HSIs, 8 (13%) re-injuries occurred within 1 month of RTS and 11 (18%) re-injuries occurred within 3 months. There were no statistically significant differences between those who did and did not re-injure in maximum eccentric force of either limb (p-values $\geq$ 0.52), total force from both limbs (p-values $\geq$ 0.47), and between-limb force asymmetry (p-values $\geq$ 0.43), regardless if re-injury occurred within 1 or 3 months after RTS.

Eccentric hamstring strength asymmetries according to injury severity, measured by days to RTS, are given in Table 3. Of the 60 index injuries analyzed, 6 (10%) were mild, 38 (63%) were moderate, and 16 (27%) were severe; no HSIs were classified as minimal. Within 1 month of RTS, no re-injuries were observed in the athletes with mild or severe index HSI, while 8 (21%) of the athletes with moderate index HSI experienced a re-injury. Within 3 months of RTS, 10 (26%) of the athletes with a moderate index HSI and 1 (6%) of the athletes with a severe index HSI were re-injured. Within the moderate index HSI group, there was no difference between those who did and did not sustain a re-injury within 1 month (p = 0.49) or 3 months (p = 0.48) after RTS. The analyses were not performed with the mild and severe index HSI groups due to low re-injury numbers.

4. Discussion

The primary purpose of this study was to determine if there were differences in eccentric hamstring strength and asymmetry at RTS between athletes who did and did not go on to sustain a re-injury. Inconsistent with our hypothesis, there was no statistically significant difference in eccentric hamstring strength asymmetry at RTS between athletes who re-injured and those who did not within 1 month or within 3 months of RTS.

Of the 60 index HSI, 13% of the athletes experienced re-injury to the same limb within the initial 1 month and 18% within the initial 3 months. Re-injury rates vary between 13.9% and 63.3% in the literature (Bodendorfer et al., 2021; De Visser et al., 2012; Ekstrand et al., 2016; Green et al., 2020). The re-injury rates of the present study are closer to the lower limit of the re-injury rates in the literature. The different time periods over which re-injuries were defined may have contributed to this range of previously reported injury rates. Many studies assess re-injuries within the initial 1 month after RTS, while others extend the follow-up period to 1 year (Bodendorfer et al., 2021; De Visser et al., 2012). The longest follow-up period for HSI to date was 2 years after index HSI, with an observed re-injury rate of 35% (Bodendorfer et al., 2021). In the literature, there is not a standard definition of the period over which to observe re-injury. Because of this, and the potential for the effect of RTS strength on re-injury to change as time from RTS increases, we opted to assess re-injury over two time intervals 1 and 3 months from RTS.

The relationship between eccentric hamstring strength and index HSI is controversial in the literature (Green et al., 2020; Hickey et al., 2017; Opar et al., 2013, 2021; Van Dyk et al., 2017; Wille et al., 2022). While early publications indicate that eccentric strength is related to hamstring injury occurrence (Opar et al., 2013, 2015), larger studies in recent years have found no association between eccentric strength and injury (Opar et al., 2021; Wille et al., 2022). For example, a recent study with 326 athletes showed that preseason hamstring strength and asymmetry measurements were not related to index HSI (Wille et al., 2022). Additional studies have observed no differences in eccentric hamstring strength between athletes with and without a history of HSI, or between preseason and RTS assessments in those who incur an HSI (Smith et al., 2022; Wangensteen et al., 2019). Our study builds upon this literature, as we found that hamstring eccentric strength magnitude and asymmetry measured at the time of RTS did not differ between those who did and did not sustain a re-injury.

The secondary purpose was to describe eccentric hamstring

Table 1
Athlete characteristics. Values represent mean ± SD unless otherwise noted.

Participant Information			
	N = 60[a]	Re-injured (N = 11)	Not Re-injured (N = 49)
Age (years)	20.2 ± 1.4	19.8 ± 1.7	20.3 ± 1.3
Height (cm)	183.3 ± 12.3	188.6 ± 19.0	182.1 ± 10.1
Mass (kg)	90.8 ± 24.0	85.9 ± 11.1	91.9 ± 26.0
Body mass index (kg/m^2)	26.8 ± 5.6	24.7 ± 4.9	27.3 ± 5.6
Females, n (% of total participants)	11 (18%)	0 (0%)	11 (22%)
Days from index HSI to RTS	22.8 ± 14.0	22.2 ± 11.7	22.9 ± 14.6
Days between RTS and testing date	6.8 ± 7.8	3.8 ± 8.4	7.5 ± 7.5
Timing of RTS, n (% of total participants)			
In-season	41 (68%)	8 (73%)	33 (67%)
Out-of-season	19 (32%)	3 (27%)	16 (33%)
Sports, n (% of total participants)			
American Football	35 (58%)	6 (55%)	29 (59%)
Track	21 (35%)	5 (45%)	16 (33%)
Soccer	4 (7%)	0 (0%)	4 (8%)

[a] HSI = hamstring strain injury, RTS = return-to-sport.

Table 2
Hamstring eccentric strength of athletes with and without a hamstring strain re-injury within the initial 1 month and initial 3 months after return-to-sport (RTS) from an index hamstring strain injury.

	1 Month[a]			3 Months[a]		
	Re-injury		p	Re-injury		p
	No	Yes		No	Yes	
Number of Athletes	52 (87%)	8 (13%)	–	49 (82%)	11 (18%)	–
Maximum Force (N)						
Injured Limb	389.2 (313.5–491.2)	376.8 (337.8–406.5)	0.75	380.2 (315.0–495.5)	386.3 (332.0–410.0)	0.54
Uninjured Limb	401.4 (318.0–486.7)	387.5 (323.0–430.5)	0.63	399.5 (322.0–501.0)	401.8 (310.0–434.6)	0.52
Total Force	810.1 (632.7–976.4)	764.3 (674.7–803.5)	0.57	796.6 (633.4–1000.8)	769.0 (647.0–839.5)	0.47
Asymmetry (%)	8.2 (5.2–13.0)	7.8 (3.9–19.5)	0.91	8.3 (4.9–13.3)	7.1 (5.3–13.3)	0.96

[a] Values reported as median and Inter-Quartile Range (25%–75%) unless indicated. p: Mann-Whitney *U* Test.

Table 3
Between limb asymmetries in hamstring eccentric strength of athletes according to injury severity.

Injury Severity		1 Month			3 Months		
		Re-injury		p	Re-injury		P
		No	Yes		No	Yes	
Mild (n = 6)	N (% of group)	6 (100%)	0 (0%)	–	6 (100%)	0 (0%)	–
	Asymmetry (%)	8.3 (1.9–13.9)	–		8.3 (1.9–13.9)	–	
Moderate (n = 38)	N (% of group)	30 (79%)	8 (21%)	0.89	28 (74%)	10 (26%)	0.79
	Asymmetry (%)	8.1 (5.9–11.2)	7.8 (3.9–19.5)		8.2 (5.7–12.3)	6.9 (4.8–15.3)	
Severe (n = 16)	N (% of group)	16 (100%)	0 (0%)	–	15 (94%)	1 (6%)	–
	Asymmetry (%)	9.3 (4.3–15.2)	–		8.8 (4.2–15.6)	9.8 –	

*Values reported as median and Inter-Quartile Range (25%–75%) unless indicated. p: Mann-Whitney *U* Test. Comparisons were not made for mild and severe injury groups due to insufficient number of re-injury.

strength asymmetry and re-injury according to the index injury severity of the athletes. Re-injuries occurred within both 1 and 3 months following RTS in athletes with moderate or severe index HSI, with no re-injuries observed in the mild index HSI group. In our study sample, 63% of all index injuries were moderate severity injuries and this group demonstrated the greatest percentage of re-injury. However, there was no difference in terms of eccentric strength or asymmetry between those within the moderate severity subgroup who did and did not re-injure. A comparison could not be made in mild and severe severity groups because of overall low re-injury numbers (0 and 1 athletes, respectively). While the greatest percentage of re-injury was seen in those who returned to sports between 2 and 4 weeks, 6% re-injury was seen by the 3rd month in those who returned to sports after the 4th week. In a recent study on hamstring re-injury in the literature, it was observed that the re-injury rate was higher in the athletes who returned to sports in the 2 nd week, and the re-injury rate decreased as the weeks progressed (Bodendorfer et al., 2021), which are similar to the trends observed in the present study.

In this study, the relationship between eccentric hamstring strength and asymmetry measured only at RTS and re-injury was investigated. Similar to the many theorized risk factors of HSIs, there are likely other risk factors that may influence re-injury occurrence. Time between RTS and eccentric hamstring strength testing may have influenced our findings as the strength test was typically performed within 7 days of RTS but ranged due to athlete readiness and scheduling availability. Athletes were also at different points in their seasons, which can influence re-injury risk. In addition, a comprehensive history of HSI in the athletes was not available, with the potential for some athletes to have had one or more prior HSIs. We attempted to minimize this influence by excluding athletes with any time loss injuries, e.g. HSI, within the 3 months prior to eccentric strength testing. Finally, the training programs for athletes following RTS may have included hamstring strengthening; however, the use of such training programs would likely have been consistent over the surveillance period and across athletes due to no changes in coaching or conditioning staff.

5. Conclusion

This study showed that eccentric hamstring strength measured at the time of RTS did not differ between those who did and did not re-injure within 3 months of RTS. Eccentric hamstring strength can be used to monitor the strength development of injured athletes. However, eccentric hamstring strength and asymmetry determined at the time of RTS after HSI does not show a distinctive difference in athletes with and without re-injury.

Ethical approval

The study was approved by the University of Wisconsin-Madison Health Sciences Institutional Review Board (IRB: 2017-0152).

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Declaration of competing interest

None.

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